PCEOT503: POWER SYSTEMS

Last-Minute High-Yield Exam Booster & Formula Cheatsheet

Master Summary of Essential Definitions, Derivations, Diagrams & Numerical Shortcuts · Modules 1 & 2

1 MODULE 1: Power Generation Technologies, Energy Storage & Load Dynamics

1.1 1. Conventional & Renewable Power Plants (Must-Know Functions & Layouts)

Plant Type	Core Principle / Thermodynamic Cycle	Key Components & Their Exact Functions
Thermal (Coal)	Rankine Cycle (Overall $\eta \approx 30-38\%$)	**Economizer:** Preheats feed-water using flue gas waste heat; **Superheater:** Converts wet steam to dry superheated steam; **Air Preheater:** Recovers flue heat to preheat combustion air; **Electrostatic Precipitator (ESP):** Traps fly ash particles.
Gas Turbine	Brayton Cycle ($\eta \approx 30-35\%$; Combined Cycle CCGT $\approx 55-60\%$)	Compressor → Combustion Chamber → Gas Turbine; CCGT pairs Gas Turbine exhaust with Heat Recovery Steam Generator (HRSG) and Steam Turbine.
Nuclear	Controlled Nuclear Fission ($^{235}\text{U}+n \rightarrow \text{Fission}+2.5n+200\,\text{MeV}$)	**Moderator:** Slows fast neutrons (D_2O, Graphite); **Control Rods:** Absorbs neutrons (Cadmium/Boron); **Coolant:** Transfers fission heat; **Shielding:** Lead/thick concrete.
Hydroelectric	Potential → Kinetic → Mechanical ($P = 9.81 \cdot \eta \cdot Q \cdot H$ kW)	**Surge Tank:** Absorbs water hammer pressure transients caused by rapid governor valve closure; **Penstock:** High-pressure conduit; **Draft Tube:** Recovers kinetic head.
Solar PV	Photovoltaic Effect ($E_{\text{photon}} \geq E_g$)	Cell → Module → Array; **MPPT (Maximum Power Point Tracking):** Perturb & Observe algorithms lock into maximum power operating point.
Wind Energy	Kinetic power $P = \frac{1}{2}\rho A v^3 C_p$	**Betz Limit:** Maximum theoretical aerodynamic power coefficient $C_{p,\max} = \frac{16}{27} \approx 59.3\%$. Pitch control & Yaw mechanism.

1.2 2. Grid Energy Storage Systems (BESS, PHES, Flywheels, Supercapacitors)

- **Battery Energy Storage (BESS):** Lithium-Ion (high energy density, 85–95% round-trip efficiency), Lead-Acid, Flow Batteries. Fast sub-second response for frequency regulation.
- **Pumped Hydro (PHES):** Largest grid storage ($> 90\%$ global); round-trip efficiency $\eta \approx 70-80\%$; pumps water to upper reservoir during off-peak and generates during peak.
- **Flywheels & Supercapacitors:** High power density, fast charge/discharge (milliseconds), millions of cycles; ideal for grid inertia and momentary voltage sag support.

1.3 3. Load Dynamics & Economic Dispatch (All Numerical Formulas)

Master Formulas for Load Curves & Plant Economics

1. Connected Load (CL) = $\sum$ Rated capacities of all connected loads
2. Maximum Demand (MD) = Peak instantaneous power demand
3. Demand Factor $= \frac{\text{Maximum Demand}}{\text{Connected Load}} \leq 1$
4. Average Load $= \frac{\text{Total Energy Generated in Time } T}{\text{Time Duration } T} = \frac{\text{Area under Load Curve}}{24 \text{ hours}}$
5. Load Factor (LF) $= \frac{\text{Average Load}}{\text{Maximum Demand}} = \frac{\text{Energy Generated in } T}{\text{MD} \times T} \leq 1$

1. Diversity Factor $= \frac{\sum \text{Individual Maximum Demands}}{\text{Coincident System Maximum Demand}} \geq 1$
2. Capacity Factor (CF) $= \frac{\text{Average Load}}{\text{Plant Rated Capacity}} = \text{LF} \times \frac{\text{MD}}{\text{Capacity}}$
3. Plant Use Factor (UF) $= \frac{\text{Energy Generated in } T}{\text{Plant Capacity} \times \text{Operating Hours } t_{\text{op}}}$
4. Reserve Capacity = Plant Rated Capacity − Maximum Demand
5. Straight Line Depreciation: $D = \frac{P-S}{n}$ [Rs/year]
6. Sinking Fund: $q = (P-S)\left[\frac{r}{(1+r)^n - 1}\right]$ [Rs/year]

Key Exam Traps & Quick Facts for Module 1

• **Diversity Factor is ALWAYS ≥ 1!** Higher diversity factor decreases coincident maximum demand, reducing required capital plant capacity and lowering generation cost per unit.
• **Base Load vs Peak Load Plants:** Base load (Coal, Nuclear, Run-of-river Hydro) have high capital cost, low running cost, and run at 100% capacity. Peak load (Gas Turbine, Pumped Storage, Diesel) have low capital cost, high running cost, and quick startup time.

2 MODULE 2: Transmission Line Parameters, Modelling, Operational Phenomena & Insulators

2.1 1. Transmission Line Inductance & Capacitance (Formula Summary)

Inductance and Capacitance Exact Working Formulas

1. Self GMR (D_s) and Internal Inductance:

- Solid Conductor Internal Inductance: $L_{\text{int}} = \frac{\mu_0}{8\pi} = \frac{1}{2} \times 10^{-7}$ H/m **(Independent of conductor radius!)**
- Fictitious Self GMR for Inductance: $D_s = r' = r \cdot e^{-1/4} = 0.7788 \cdot r$ (Accounts for internal flux linkages).
- For Capacitance: **ALWAYS USE PHYSICAL RADIUS r** (Electrostatic charge resides exclusively on outer surface).

2. 3-Phase Single Circuit Lines (Transposed):

$$D_m = \sqrt[3]{D_{12}D_{23}D_{31}}, \qquad L = 2 \times 10^{-7} \ln\left(\frac{D_m}{D_s}\right) \text{ H/m/phase}, \qquad C_n = \frac{2\pi\epsilon_0}{\ln(D_m/r)} \text{ F/m/phase}$$

3. 3-Phase Double Circuit Lines (Inverted Hexagonal $a-b-c$ vs $c'-b'-a'$):

$$D_{sa} = \sqrt{r'D_{aa'}}, \quad D_{sb} = \sqrt{r'D_{bb'}}, \quad D_{sc} = \sqrt{r'D_{cc'}} \implies D_s = \sqrt[3]{D_{sa}D_{sb}D_{sc}}$$

$$D_{ab} = \sqrt[4]{D_{ab}D_{ab'}D_{a'b}D_{a'b'}}, \quad D_{bc} = \sqrt[4]{D_{bc}D_{bc'}D_{b'c}D_{b'c'}}, \quad D_{ca} = \sqrt[4]{D_{ca}D_{ca'}D_{c'a}D_{c'a'}} \implies D_m = \sqrt[3]{D_{ab}D_{bc}D_{ca}}$$

$$L = 2 \times 10^{-7} \ln\left(\frac{D_m}{D_s}\right) \text{ H/m/phase}, \qquad C_n = \frac{2\pi\epsilon_0}{\ln(D_m/D_{sc})} \text{ F/m/phase} \quad (\text{with } D_{sca} = \sqrt{rD_{aa'}})$$

2.2 2. Transmission Line Modelling & Two-Port ABCD Parameters

Model	Length / Volt	A	B	C	D
Short Line	< 80 km, $<$ 20 kV	1	Z	0	1
Nominal-T	$80 - 250$ km	$1 + \frac{YZ}{2}$	$Z\left(1 + \frac{YZ}{4}\right)$	Y	$1 + \frac{YZ}{2}$
Nominal-π	$80 - 250$ km	$1 + \frac{YZ}{2}$	Z	$Y\left(1 + \frac{YZ}{4}\right)$	$1 + \frac{YZ}{2}$
Long Line	> 250 km	$\cosh(\gamma l)$	$Z_c \sinh(\gamma l)$	$\frac{1}{Z_c}\sinh(\gamma l)$	$\cosh(\gamma l)$

- **Fundamental Properties:** Symmetric $\iff A = D$; Reciprocal $\iff AD - BC = 1$.
- **Sending End Matrix:** $V_s = AV_r + BI_r, \quad I_s = CV_r + DI_r$.
- **Percentage Voltage Regulation:** $\%VR = \frac{|V_s/A| - |V_r|}{|V_r|} \times 100\%$ **(Do NOT forget to divide V_s by A for medium/long lines!).**
- **Transmission Efficiency:** $\eta = \frac{P_r}{P_s} \times 100\% = \frac{3|V_{r,\text{ph}}||I_r|\cos\phi_r}{3\,\text{Re}(V_s I_s^*)} \times 100\%$.

2.3 3. Operational Phenomena (Key Exam Definitions & Derivations)

Phenomenon	Core Definition / Cause	Mathematical Expression	Consequences & Mitigation
Skin Effect	Concentration of AC current near conductor outer periphery due to internal flux eddy currents	Skin Depth $\delta = \sqrt{\frac{\rho}{\pi f \mu}}$	Increases effective resistance ($R_{\text{ac}} > R_{\text{dc}}$). Mitigated by stranded ACSR conductors.
Ferranti Effect	No-load / light-load receiving end voltage exceeds sending end voltage ($V_r > V_s$) due to capacitive charging current	$\Delta V = V_r - V_s \approx \frac{V_r \omega^2 l^2 LC}{2} > 0$	Overvoltage stress on insulation; mitigated by connecting **Shunt Reactors** at receiving end.
Corona	Dielectric breakdown of air surrounding conductor when surface gradient exceeds $g_0 = 21.2$ kV/cm (rms)	$V_d = m_0 g_0 \delta r \ln(d/r)$ $P = \frac{242.2}{\delta}(f+25)\sqrt{\frac{r}{d}}(V - V_d)^2 \times 10^{-5}$	Pale-violet glow, hissing noise, O_3 corrosion, power loss. Mitigated by bundled conductors.
Surge Impedance Loading (SIL)	Power delivered to purely resistive load equal to surge impedance $Z_c = \sqrt{L/C}$	$\text{SIL} = \frac{V_{L\text{-}L}^2}{Z_c}$ ($Z_c \approx 400\,\Omega$ overhead, $40\,\Omega$ cable)	At SIL, $Q_C = Q_L$, flat voltage profile ($V_s = V_r$). $P <$ SIL $\rightarrow$ voltage rise; $P >$ SIL $\rightarrow$ voltage drop.

2.4 4. Insulators & String Efficiency (Quick Formula Guide)

- **Disc Voltages (3-Disc String with shunt ratio $k = C_1/C$):**

$$V_2 = V_1(1+k), \qquad V_3 = V_1(1+3k+k^2), \qquad V = V_1(3+4k+k^2)$$

- **Maximum Voltage Stress:** Occurs across the disc **nearest to the conductor** (V_3).
- **String Efficiency:** $\eta = \frac{V}{n \times V_n} \times 100\%$.
- **Methods of Improvement:** Longer cross-arms (reduces k), Capacitance Grading ($C_1 < C_2 < C_3$), Guard Ring (injects current from line to achieve 100% uniform voltage distribution).